Name _____ Period _____ Date _____

2-4 Lesson Quiz: Complex Numbers and Operations short form

Questions 2, 3 and 4 of the Savvas 2-4 Lesson Quiz, numbered as on the Savvas form. Show your work.

2. What is the difference of $(5 + 3i)$ and $(2 + 9i)$?

Ⓐ $7 + 12i$

Ⓑ $3 + 6i$

Ⓒ $3 - 6i$

Ⓓ $7 + 6i$

Work:

3. Write the product $(2+7i)(2-7i)$ in the form $a+bi$.

Ⓐ $4 - 49i$

Ⓑ 53

Ⓒ $4 - 14i^2$

Ⓓ $53 + 14i$

Work:

4. Write the expression $3i(2 - 5i)(4 + i)$ in the form $a + bi$.

Work:

_____ + _____ i

Scoring: Question 3 and Question 4 are 10 points each. Question 2 is a bonus of 5 points. Total out of 20.

2-4 Lesson Quiz, short form: answer key

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2. **C,** $3-6i$**.** $(5+3i)-(2+9i)=(5+3i)+(-2-9i)=(5-2)+(3i-9i)=3-6i$. A adds the two numbers; B subtracts the imaginary parts in the wrong order ($9i-3i$); D adds the real parts and makes the same imaginary-part error. Savvas quiz item 2, DOK 1, Skills Review NS11 (taught by Example 2, cut from the fast packet; Practice b rehearses it). **Bonus 5:** 5 for C with the subtraction shown; 3 for C with no work; 0 otherwise.

3. **B,** 53**.** $(2+7i)(2-7i)=4-14i+14i-49i^2=4-49(-1)=53$. The factors are complex conjugates, so the product is $2^2+7^2=53$, a real number. A writes $-49i^2$ as $-49i$; C and D mishandle the middle terms or i^2. Savvas quiz item 3, DOK 1, Skills Review NS11 (Example 3B, Try It 3b). **10 points:** 7 for B, 3 for work that shows $i^2=-1$ used.

4. $54+39i$ (blanks: 54; 39).
$(2-5i)(4+i)=8+2i-20i-5i^2=8-18i+5=13-18i$, then $3i(13-18i)=39i-54i^2=54+39i$.
Other order, same answer: $3i(2-5i)=15+6i$, then $(15+6i)(4+i)=60+15i+24i+6i^2=54+39i$.
Savvas quiz item 4, DOK 2, Skills Review NS11 (Example 3A and 3B; Practice a is the parallel). **10 points:** 4 for the first product, 4 for the second product with $i^2=-1$, 2 for the answer written as $a+bi$.

Items left off this form (available for a retake or Form B): Savvas item 1, solve $x^2=-64$ over the complex numbers, select all (answer A and D, $8i$ and $-8i$; DOK 1; Example 1, cut); Savvas item 5, match each product with its factors (answers: $18x^2+2$ is 2, $(3x-i)$, $(3x+i)$; $3x^2+12$ is 3, $(x-2i)$, $(x+2i)$; $36x^2+25$ is $(6x-5i)(6x+5i)$; $9x^2+1$ is $(3x-i)(3x+i)$; DOK 2; Example 5, cut). Both use the Skills Review code NS11. The TE prints no standards codes; the curriculum map's Topic 2 list gives HSN.CN.A.1 and HSN.CN.A.2 for this lesson's operations.

IXL replacement: SmartScore 70 or above on each starred skill, VZ8, 7U5 and JVF, replaces this quiz with 20/20.